Лабораторная работа №4

Работа с программными пакетами

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2025

Докладчик

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Цель работы

- Получить навыки работы с репозиториями и менеджерами пакетов
- Изучить основные команды для управления пакетами в Linux
- Научиться работать с rpm-пакетами

Теоретическая справка

Управление пакетами в Linux

Основные понятия: - `dnf` — менеджер пакетов в Rocky Linux - `rpm` — низкоуровневый менеджер пакетов - Репозитории — источники пакетов - Группы пакетов — наборы связанных пакетов

Режим работы суперпользователя

```
root@vbox:~# rpm -ql dnsmasq
/etc/dbus-1/system.d/dnsmasq.conf
/etc/dnsmasq.conf
/etc/dnsmasq.d
/usr/lib/.build-id
/usr/lib/.build-id/c9
/usr/lib/.build-id/c9/a4bb4576977dbef150b40955510f869d246b34
/usr/lib/systemd/system/dnsmasq.service
/usr/lib/sysusers.d/dnsmasq.conf
/usr/sbin/dnsmasq
/usr/share/dnsmasq
/usr/share/dnsmasq/trust-anchors.conf
/usr/share/doc/dnsmasq
/usr/share/doc/dnsmasq/CHANGELOG
/usr/share/doc/dnsmasq/DBus-interface
/usr/share/doc/dnsmasq/FAQ
```

Рис. 1: Рис. 1. Открытие режима работы суперпользователя и последующее открытие каталога

Содержание файла rocky addons.repo

```
root@vbox:~# dnf list dnsmasq
Última verificação de metadados: 0:28:40 atrás em qua 28 jan 2026 18:05:37.
Pacotes instalados
dnsmasq.x86_64                          2.90-4.el10                          @AppStream
root@vbox:~# dnf install dnsmasq
Última verificação de metadados: 0:29:18 atrás em qua 28 jan 2026 18:05:37.
O pacote dnsmasq-2.90-4.el10.x86_64 já está instalado.
```

Рис. 2: Рис. 2. Содержание файла cat rocky addons.repo

Содержание файла rocky-devel.repo

```
root@vbox:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# rpm -e lynx
root@vbox:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# ls
lynx-2.9.0-6.el10.x86_64.rpm
```

Рис. 3: Рис. 3. Содержание файла cat rocky-devel.repo

Содержание файла rocky-extras.repo

```
                                                    Welcome to Rocky Linux (p1 of 2)
  Rocky Linux 10

                                Welcome to Rocky Linux

A community Enterprise Operating System

    ---------------------------------------------------------------------

  Rocky Linux is an Enterprise Linux distribution based on the sources
  available from the CentOS Project and Red Hat, Inc. used for CentOS Stream
  and Red Hat Enterprise Linux. Rocky Linux intends to be binary compatible
  with the upstream distributions. The source code can be found at our Rocky
  Linux Git Service.

  Rocky Linux is developed by a diverse team of core developers, with various
  backgrounds and skill sets that allow them to build, test, and provide
  Rocky Linux to the world. The core developers are supported by an active
  user community as well, which includes many testers from the Rocky Linux
```

Рис. 4: Рис. 4. Содержание файла cat rocky-extras.repo

Содержание файла rocky.repo

```
                                                    Welcome to Rocky Linux (p1 of 2)
   Rocky Linux 10

                                Welcome to Rocky Linux

A community Enterprise Operating System

     ----------------------------------------------------------------------

   Rocky Linux is an Enterprise Linux distribution based on the sources
   available from the CentOS Project and Red Hat, Inc. used for CentOS Stream
   and Red Hat Enterprise Linux. Rocky Linux intends to be binary compatible
   with the upstream distributions. The source code can be found at our Rocky
   Linux Git Service.

   Rocky Linux is developed by a diverse team of core developers, with various
   backgrounds and skill sets that allow them to build, test, and provide
   Rocky Linux to the world. The core developers are supported by an active
   user community as well, which includes many testers from the Rocky Linux
```

Рис. 5: Рис. 5. Содержание файла cat rocky.repo

Список репозиториев и пакетов

```
root@vbox:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# rpm -qc lynx
/etc/lynx-site.cfg
/etc/lynx.cfg
/etc/lynx.lss
```

Рис. 6: Рис. 6. Список репозиториев и пакетов

Поиск nmap

```
/usr/share/doc/lynx/README
/usr/share/doc/lynx/docs/CHANGES2.3
/usr/share/doc/lynx/docs/CHANGES2.4
/usr/share/doc/lynx/docs/CHANGES2.5
/usr/share/doc/lynx/docs/CHANGES2.6
/usr/share/doc/lynx/docs/CHANGES2.7
/usr/share/doc/lynx/docs/CHANGES2.8
/usr/share/doc/lynx/docs/CRAWL.announce
/usr/share/doc/lynx/docs/FM.announce
/usr/share/doc/lynx/docs/IBMPC-charsets.announce
/usr/share/doc/lynx/docs/README.TRST
/usr/share/doc/lynx/docs/README.chartrans
/usr/share/doc/lynx/docs/README.cookies
/usr/share/doc/lynx/docs/README.defines
/usr/share/doc/lynx/docs/README.metrics
/usr/share/doc/lynx/docs/README.options
```

Рис. 7: Рис. 7. Выполнение команды dnf search nmap

Информация о nmap

```
root@vbox:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# rpm -qd lynx
/usr/share/doc/lynx/README
/usr/share/doc/lynx/docs/CHANGES2.3
/usr/share/doc/lynx/docs/CHANGES2.4
/usr/share/doc/lynx/docs/CHANGES2.5
/usr/share/doc/lynx/docs/CHANGES2.6
/usr/share/doc/lynx/docs/CHANGES2.7
/usr/share/doc/lynx/docs/CHANGES2.8
/usr/share/doc/lynx/docs/CRAWL.announce
/usr/share/doc/lynx/docs/FM.announce
/usr/share/doc/lynx/docs/IBMPC-charsets.announce
/usr/share/doc/lynx/docs/README.TRST
/usr/share/doc/lynx/docs/README.chartrans
/usr/share/doc/lynx/docs/README.cookies
/usr/share/doc/lynx/docs/README.defines
/usr/share/doc/lynx/docs/README.metrics
/usr/share/doc/lynx/docs/README.options
/usr/share/doc/lynx/docs/README.release
/usr/share/doc/lynx/docs/README.rootcerts
```

Рис. 8: Рис. 8. Выполнение команды dnf info nmap

Установка nmap

```
root@vbox:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# rpm -qd lynx
/usr/share/doc/lynx/README
/usr/share/doc/lynx/docs/CHANGES2.3
/usr/share/doc/lynx/docs/CHANGES2.4
/usr/share/doc/lynx/docs/CHANGES2.5
/usr/share/doc/lynx/docs/CHANGES2.6
/usr/share/doc/lynx/docs/CHANGES2.7
/usr/share/doc/lynx/docs/CHANGES2.8
/usr/share/doc/lynx/docs/CRAWL.announce
/usr/share/doc/lynx/docs/FM.announce
/usr/share/doc/lynx/docs/IBMPC-charsets.announce
/usr/share/doc/lynx/docs/README.TRST
/usr/share/doc/lynx/docs/README.chartrans
/usr/share/doc/lynx/docs/README.cookies
/usr/share/doc/lynx/docs/README.defines
/usr/share/doc/lynx/docs/README.metrics
/usr/share/doc/lynx/docs/README.options
/usr/share/doc/lynx/docs/README.release
/usr/share/doc/lynx/docs/README.rootcerts
```

Рис. 9: Рис. 9. Выполнение команды dnf install nmap

Установка nmap*

```
root@vbox:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# rpm -qi lynx
Name        : lynx
Version     : 2.9.0
Release     : 6.el10
Architecture: x86_64
Install Date: qua 28 jan 2026 18:13:35
Group       : Unspecified
Size        : 6283241
License     : GPL-2.0-only
Signature   :
              RSA/SHA256, qui 22 mai 2025 23:34:29, Key ID 5b106c736fedfc85
Source RPM  : lynx-2.9.0-6.el10.src.rpm
Build Date  : sex 01 nov 2024 13:03:33
Build Host  : pb-913aa2bb-e2a2-4928-a48c-b613e15933c9-b-x86-64
Packager    : Rocky Linux Build System (Peridot) <releng@rockylinux.org>
Vendor      : Rocky Enterprise Software Foundation
```

Рис. 10: Рис. 10. Выполнение команды dnf install nmap*

Удаление nmap

```
root@vbox:/etc/yum.repos.d# find /var/cache/dnf/ -name lynx
root@vbox:/etc/yum.repos.d# find /var/cache/dnf/ -name lynx*
/var/cache/dnf/appstream-8f18e4146890fdfc/packages/lynx-2.9.0-6.el10.x86_64.rpm
root@vbox:/etc/yum.repos.d# cd /var/cache/dnf/appstrem-8f18e4146890fdfc/packages
-bash: cd: /var/cache/dnf/appstrem-8f18e4146890fdfc/packages: Arquivo ou diretório i
nexistente
root@vbox:/etc/yum.repos.d# cd /var/cache/dnf/appstream-8f18e4146890fdfc/packages
root@vbox:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# ls
lynx-2.9.0-6.el10.x86_64.rpm
root@vbox:/var/cache/dnf/appstream-8f18e4146890fdfc/packages# rpm -Uhv lynx-2.9.0-6.
el10.x86_64.rpm
Verificando...                          ################################# [100%]
```

Рис. 11: Рис. 11. Выполнение команды dnf remove nmap

Удаление nmap*

```
root@vbox:/etc/yum.repos.d# dnf install lynx --downloadonly
Última verificação de metadados: 0:00:13 atrás em qua 28 jan 2026 18:05:37.
Dependências resolvidas.
================================================================================
 Pacote         Arquitetura      Versão                   Repositório        Tam.
================================================================================
Instalando:
 lynx           x86_64           2.9.0-6.el10             appstream          1.6 M

Resumo da transação
================================================================================
Instalar  1 pacote

Tamanho total do download: 1.6 M
Tamanho depois de instalado: 6.0 M
DNF fará o download apenas dos pacotes para a transação.
Correto? [s/N]: s
```

Рис. 12: Рис. 12. Выполнение команды dnf remove nmap*

Список групп пакетов

```
root@vbox:/etc/yum.repos.d# dnf list lynx
Última verificação de metadados: 4:08:59 atrás em qua 28
Pacotes disponíveis
lynx.x86_64                                2.9.0-6.el10
```

Рис. 13: Рис. 13. Получение списков имеющихся групп пакетов (выполнение команды dnf groups list)

Список групп пакетов (LANG=C)

```
root@vbox:/etc/yum.repos.d# dnf history
ID     | Linha de comando              | Data e hora      | Ação(ões)      | Alterad
-------------------------------------------------------------------------------------
     9 | groupremove RPM Development t | 28-01-2026 18:02 | Removed        |   24
     8 | groupinstall RPM Development  | 28-01-2026 18:01 | Install        |   25
```

Рис. 14: Рис. 14. Выполнение команды LANG=C dnf group list

Информация о группе RPM Development Tools

```
root@vbox:/etc/yum.repos.d# dnf groupremove "RPM Development tools"
Dependências resolvidas.
================================================================================
 Pacote                      Arq.     Versão                    Repositório   Tam.
================================================================================
Removendo:
 redhat-rpm-config           noarch   293-1.el10.rocky.0.2      @appstream   180 k
 rpm-build                   x86_64   4.19.1.1-20.el10          @appstream   173 k
 rpmdevtools                 noarch   9.6-9.el10                @appstream   198 k
Removendo dependências não utilizadas:
```

Рис. 15: Рис. 15. Получение информации (выполнение команды dnf groups info “RPM Development Tools”)

Установка группы RPM Development Tools

```
root@vbox:/etc/yum.repos.d# dnf groupinstall "RPM Development tools"
Última verificação de metadados: 4:05:06 atrás em qua 28 jan 2026 13:55:39.
Dependências resolvidas.
================================================================================
 Pacote                    Arq.     Versão                    Repositório  Tam.
================================================================================
Instalando grupo/pacotes do módulo:
 redhat-rpm-config         noarch   293-1.el10.rocky.0.2      appstream    78 k
 rpm-build                 x86_64   4.19.1.1-20.el10          appstream    76 k
 rpmdevtools               noarch   9.6-9.el10                appstream    87 k
Instalando dependências:
 debugedit                 x86_64   5.1-8.el10                appstream    79 k
 dwz                       x86_64   0.16-1.el10               appstream   139 k
 efi-srpm-macros           noarch   6-7.el10                  appstream    23 k
 elfutils                  x86_64   0.193-1.el10.rocky.0.1    baseos      567 k
```

Рис. 16: Рис. 16. Установка группы пакетов RPM Development Tools (выполнение команды dnf groupinstall “RPM Development Tools”)

Удаление группы RPM Development Tools

```
root@vbox:/etc/yum.repos.d# dnf groups info "RPM Development tools"
Última verificação de metadados: 4:04:03 atrás em qua 28 jan 2026 13:55:39.
Grupo: Ferramentas de Desenvolvimento RPM
```

Рис. 17: Рис. 17. Удаление группы пакетов RPM Development Tools (выполнение команды dnf groupremove “RPM Development Tools”)

Просмотр истории dnf

```
root@vbox:/etc/yum.repos.d# dnf install nmap\*
Última verificação de metadados: 3:58:59 atrás em qua 28
O pacote nmap-4:7.92-3.el10.x86_64 já está instalado.
O pacote nmap-ncat-4:7.92-3.el10.x86_64 já está instalado
Dependências resolvidas.
Nada para fazer.
Concluído!
```

Рис. 18: Рис. 18. Просмотр использования команды dnf и отмена шестого по счёту действия

Скачивание пакета lynx

```
root@vbox:/etc/yum.repos.d# dnf groups list
Última verificação de metadados: 4:01:35 atrás em qua 28 jan 2026 13:55:39.
Grupos de Ambientes Disponíveis:
   Server
   Instalação Mínima
   Workstation
   Custom Operating System
   Virtualization Host
Grupos de ambientes instalados:
   Server with GUI
Grupos instalados:
   Gerenciamento de contêiner
   Gerenciamento por linha de comando
Grupos disponíveis:
   Legacy UNIX Compatibility
   Smart Card Support
   Console Internet Tools
   Ferramentas de desenvolvimento
```

Рис. 19: Рис. 19. Скачивание rpm-пакета lynx

Действия с пакетом lynx

```
root@vbox:/etc/yum.repos.d# dnf remove nmap
Sem correspondência para o argumento: nmap
Nenhum pacote marcado para remoção.
Dependências resolvidas.
Nada para fazer.
Concluído!
```

Рис. 20: Рис. 20. Нахождение каталога с пакетом, установка rpm-пакета, определение расположения исполняемого файла, определение к какому пакету принадлежит lynx

Информация о содержимом пакета

```
Resumo da transação
================================================================================
Remover  2 pacotes

Espaço liberado: 24 M
Correto? [s/N]: s
Executando verificação da transação
Verificação de transação concluída.
Executando teste de transação
Teste de transação concluído.
Executando a transação
  Preparando          :                                                    1/1
  Apagando            : nmap-4:7.92-3.el10.x86_64                          1/2
  Executando scriptlet: nmap-ncat-4:7.92-3.el10.x86_64                     2/2
  Apagando            : nmap-ncat-4:7.92-3.el10.x86_64                     2/2
  Executando scriptlet: nmap-ncat-4:7.92-3.el10.x86_64                     2/2

Removido(s):
  nmap-4:7.92-3.el10.x86_64              nmap-ncat-4:7.92-3.el10.x86_64
```

Рис. 21: Рис. 21. Получение дополнительной информации о

Список всех файлов в пакете

```
root@vbox:/etc/yum.repos.d# dnf install nmap\*
Última verificação de metadados: 3:58:59 atrás em qua 28 jan 2026 13:55:39.
O pacote nmap-4:7.92-3.el10.x86_64 já está instalado.
O pacote nmap-ncat-4:7.92-3.el10.x86_64 já está instalado.
Dependências resolvidas.
Nada para fazer.
Concluído!
```

Рис. 22: Рис. 22. Получение списка всех файлов в пакете

Файлы с документацией пакета

```
root@vbox:/etc/yum.repos.d# dnf install nmap
Última verificação de metadados: 3:57:51 atrás em qua 28 jan 2026 13:55:39.
Dependências resolvidas.
================================================================================
 Pacote         Arquitetura      Versão                    Repositório        Tam.
================================================================================
Instalando:
 nmap           x86_64           4:7.92-3.el10             appstream          5.6 M

Resumo da transação
================================================================================
Instalar  1 pacote

Tamanho total do download: 5.6 M
Tamanho depois de instalado: 24 M
Correto? [s/N]: s
Baixando pacotes:
nmap-7.92-3.el10.x86_64.rpm                          1.7 MB/s | 5.6 MB     00:03
--------------------------------------------------------------------------------
```

Рис. 23: Рис. 23. Вывод перечня файлов с документацией пакета

Просмотр файлов документации

```
nmap-ncat.x86_64 : Nmap's Netcat replacement
root@vbox:/etc/yum.repos.d# dnf info  nmap
Última verificação de metadados: 3:56:09 atrás em qua 28 jan 2026 13:55:39.
Pacotes disponíveis
Nome          : nmap
Época         : 4
Versão        : 7.92
Lançamento    : 3.el10
Arquitetura   : x86_64
Tamanho       : 5.6 M
Origem        : nmap-7.92-3.el10.src.rpm
Repositório   : appstream
Resumo        : Network exploration tool and security scanner
URL           : http://nmap.org/
Licença       : LicenseRef-Nmap
Descrição     : Nmap is a utility for network exploration or security auditing.
              : It supports ping scanning (determine which hosts are up), many
              : port scanning techniques (determine what services the hosts are
              : offering), and TCP/IP fingerprinting (remote host operating
              : system identification). Nmap also offers flexible target and port
```

Конфигурационные файлы пакета

```
root@vbox:/etc/yum.repos.d# dnf search nmap
Última verificação de metadados: 3:54:54 atrás em qua 28 jan 2026 13:55:39.
========================= Nome Correspondeu Exatamente: nmap =========================
```

Рис. 25: Рис. 25. Вывод на экран перечень и месторасположение конфигурационных файлов пакета, вывод расположения и содержание скриптов

Запуск текстового браузера lynx

```
root@vbox:/etc/yum.repos.d# dnf repolist
id do repo                          nome do repo
appstream                           Rocky Linux 10 - AppStream
baseos                              Rocky Linux 10 - BaseOS
extras                              Rocky Linux 10 - Extras
root@vbox:/etc/yum.repos.d# dnf search user
Última verificação de metadados: 3:54:01 atrás em qua 28 jan 2026 13:55:39.
========================= Nome & Resumo Correspondeu: user =========================
gnome-user-docs.noarch : GNOME User Documentation
libuser.x86_64 : A user and group account administration library
perl-User-pwent.noarch : By-name interface to Perl built-in user name resolver
samba-usershares.noarch : Provides support for non-root user shares
usermode.x86_64 : Tools for certain user account management tasks
userspace-rcu.x86_64 : RCU (read-copy-update) implementation in user-space
xdg-user-dirs.x86_64 : Handles user special directories
============================== Nome Correspondeu: user ==============================
gnome-shell-extension-user-theme.noarch : Support for custom themes in GNOME
                                        : Shell
xdg-user-dirs-gtk.x86_64 : Gnome integration of special directories
============================= Resumo Correspondeu: user =============================
```

Удаление пакета и проверка

```
root@vbox:/etc/yum.repos.d# cat rocky.repo
# rocky.repo
#
# The mirrorlist system uses the connecting IP address of the client and the
# update status of each mirror to pick current mirrors that are geographically
# close to the client.  You should use this for Rocky updates unless you are
# manually picking other mirrors.
#
# If the mirrorlist does not work for you, you can try the commented out
# baseurl line instead.

[baseos]
name=Rocky Linux $releasever - BaseOS
mirrorlist=https://mirrors.rockylinux.org/mirrorlist?arch=$basearch&repo=BaseOS
$releasever$rltype
#baseurl=http://dl.rockylinux.org/$contentdir/$releasever/BaseOS/$basearch/os/
gpgcheck=1
enabled=1
countme=1
metadata_expire=6h
```

Установка пакета dnsmasq

```
root@vbox:/etc/yum.repos.d# cat rocky-extras.repo
# rocky-extras.repo
#
# The mirrorlist system uses the connecting IP address of the client and the
# update status of each mirror to pick current mirrors that are geographically
# close to the client.  You should use this for Rocky updates unless you are
# manually picking other mirrors.
#
# If the mirrorlist does not work for you, you can try the commented out
# baseurl line instead.

[extras]
name=Rocky Linux $releasever - Extras
mirrorlist=https://mirrors.rockylinux.org/mirrorlist?arch=$basearch&repo=extras
$releasever$rltype
#baseurl=http://dl.rockylinux.org/$contentdir/$releasever/extras/$basearch/os/
gpgcheck=1
enabled=1
countme=1
metadata_expire=6h
```

Информация о пакете dnsmasq

```
root@vbox:/etc/yum.repos.d# cat rocky-devel.repo
# rocky-devel.repo
#
# devel and no-package-left-behind

[devel]
name=Rocky Linux $releasever - Devel WARNING! FOR BUILDROOT ONLY DO NOT LEAVE EN
ABLED
mirrorlist=https://mirrors.rockylinux.org/mirrorlist?arch=$basearch&repo=devel-$
releasever$rltype
#baseurl=http://dl.rockylinux.org/$contentdir/$releasever/devel/$basearch/os/
gpgcheck=1
enabled=0
countme=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-Rocky-10

[devel-debuginfo]
name=Rocky Linux $releasever - Devel Debug WARNING! FOR BUILDROOT ONLY DO NOT LE
AVE ENABLED
mirrorlist=https://mirrors.rockylinux.org/mirrorlist?arch=$basearch&repo=devel-$
```

Список файлов пакета dnsmasq

```
root@vbox:/etc/yum.repos.d# cat rocky-addons.repo
# rocky-addons.repo
#
# The mirrorlist system uses the connecting IP address of the client and the
# update status of each mirror to pick current mirrors that are geographically
# close to the client.  You should use this for Rocky updates unless you are
# manually picking other mirrors.
#
# If the mirrorlist does not work for you, you can try the commented out
# baseurl line instead.

[highavailability]
name=Rocky Linux $releasever - High Availability
mirrorlist=https://mirrors.rockylinux.org/mirrorlist?arch=$basearch&repo=HighAva
ilability-$releasever$rltype
#baseurl=http://dl.rockylinux.org/$contentdir/$releasever/HighAvailability/$base
arch/os/
```

Рис. 30: Рис. 30. Получение списка всех файлов в пакете

Документация пакета dnsmasq

```
root@vbox:/etc/yum.repos.d# cat rocky-addons.repo
# rocky-addons.repo
#
# The mirrorlist system uses the connecting IP address of the client and the
# update status of each mirror to pick current mirrors that are geographically
# close to the client.  You should use this for Rocky updates unless you are
# manually picking other mirrors.
#
# If the mirrorlist does not work for you, you can try the commented out
# baseurl line instead.

[highavailability]
name=Rocky Linux $releasever - High Availability
mirrorlist=https://mirrors.rockylinux.org/mirrorlist?arch=$basearch&repo=HighAva
ilability-$releasever$rltype
#baseurl=http://dl.rockylinux.org/$contentdir/$releasever/HighAvailability/$base
arch/os/
gpgcheck=1
enabled=0
countme=1
```

Просмотр документации dnsmasq

```
root@vbox:~# cd /etc/yum.repos.d
root@vbox:/etc/yum.repos.d# ls
rocky-addons.repo  rocky-devel.repo  rocky-extras.repo  rocky.repo
root@vbox:/etc/yum.repos.d#
```

Рис. 32: Рис. 32. Просмотр файлов документации

Конфигурационные файлы dnsmasq

```
root@vbox:~# rpm -qi dnsmasq
Name        : dnsmasq
Version     : 2.90
Release     : 4.el10
Architecture: x86_64
Install Date: qua 28 jan 2026 04:59:31
Group       : Unspecified
Size        : 794976
License     : GPL-2.0-only OR GPL-3.0-only
Signature   :
              RSA/SHA256, qui 22 mai 2025 03:17:11, Key ID 5b106c736fedfc85
Source RPM  : dnsmasq-2.90-4.el10.src.rpm
Build Date  : qui 31 out 2024 14:46:28
Build Host  : pb-6df20f45-04b4-477b-bded-5a041ce1b59d-b-x86-64
Packager    : Rocky Linux Build System (Peridot) <releng@rockylinux.org>
Vendor      : Rocky Enterprise Software Foundation
VCS         : git:git://thekelleys.org.uk/dnsmasq.git
URL         : http://www.thekelleys.org.uk/dnsmasq/
```

Рис. 33: Рис. 33. Вывод на экран перечень и месторасположение конфигурационных файлов пакета

Скрипты при установке пакета

```
root@vbox:~# rpm -qd dnsmasq
/usr/share/doc/dnsmasq/CHANGELOG
/usr/share/doc/dnsmasq/DBus-interface
/usr/share/doc/dnsmasq/FAQ
/usr/share/doc/dnsmasq/doc.html
/usr/share/doc/dnsmasq/setup.html
/usr/share/man/man8/dnsmasq.8.gz
```

Рис. 34: Рис. 34. Вывод на экран расположение и содержание скриптов, выполняемых при установке пакета

Удаление пакета dnsmasq

```
root@vbox:~# dnf install dnsmasq
Última verificação de metadados: 0:33:21 atrás em qua 28
O pacote dnsmasq-2.90-4.el10.x86_64 já está instalado.
Dependências resolvidas.
Nada para fazer.
Concluído!
```

Рис. 35: Рис. 35. Удаление пакета

Проверка удаления

```
DNSMASQ(8)                    System Manager's Manual                    DNSMASQ(8)

NAME
       dnsmasq - A lightweight DHCP and caching DNS server.

SYNOPSIS
       dnsmasq [OPTION]...

DESCRIPTION
       dnsmasq  is  a  lightweight  DNS, TFTP, PXE, router advertisement and DHCP
       server. It is intended to provide coupled DNS and DHCP service to a LAN.

       Dnsmasq accepts DNS queries and either answers them from a  small,  local,
       cache or forwards them to a real, recursive, DNS server. It loads the con-
       tents  of  /etc/hosts  so  that local hostnames which do not appear in the
       global DNS can be resolved and also answers DNS queries for  DHCP  config-
       ured  hosts.  It  can  also act as the authoritative DNS server for one or
       more domains, allowing local names to appear in the global DNS. It can be
```

Рис. 36: Рис. 36. Проверка удаления пакета

Дополнительные репозитории

```
root@vbox:~# rpm -qc dnsmasq
/etc/dbus-1/system.d/dnsmasq.conf
/etc/dnsmasq.conf
```

Рис. 37: Рис. 37. Просмотр дополнительных репозиториев

Обновление кэша пакетов

```
root@vbox:~# rpm -q --scripts dnsmasq
preinstall scriptlet (using /bin/sh):
#precreate users so that rpm can install files owned by that user

# generated from dnsmasq-systemd-sysusers.conf
getent group 'dnsmasq' >/dev/null || groupadd -r 'dnsmasq' || :
getent passwd 'dnsmasq' >/dev/null || \
    useradd -r -g 'dnsmasq' -d '/var/lib/dnsmasq' -s '/usr/sbin/nologin' -c 'Dnsmasq
 DHCP and DNS server' 'dnsmasq' || :
postinstall scriptlet (using /bin/sh):


if [ $1 -eq 1 ] && [ -x "/usr/lib/systemd/systemd-update-helper" ]; then
    # Initial installation
    /usr/lib/systemd/systemd-update-helper install-system-units dnsmasq.service || :

fi
```

Рис. 38: Рис. 38. Обновление кэша пакетов

Поиск пакетов

```
root@vbox:~# rpm -e dnsmask
erro: o pacote dnsmask não está instalado
```

Рис. 39: Рис. 39. Поиск пакетов по ключевому слову

Зависимости пакетов

```
root@vbox:~# rpm -e dnsmask
erro: o pacote dnsmask não está instalado
```

Рис. 40: Рис. 40. Просмотр зависимостей пакета

Проверка целостности пакета

```
root@vbox:~# yum search useradd
Última verificação de metadados: 0:44:58 atrás em qua 28
Nenhuma correspondência encontrada.
```

Рис. 41: Рис. 41. Проверка целостности скачанного пакета

Итоговая проверка

```
root@vbox:~# yum info gcl
```

Рис. 42: Рис. 42. Итоговая проверка работы с пакетами

Вывод

В ходе выполнения лабораторной работы были получены навыки работы с репозиториями и менеджерами пакетов в операционной системе Linux. Освоены команды `dnf` для поиска, установки и удаления пакетов, работа с группами пакетов, а также управление rpm-пакетами.

Список литературы

[1] Linux man pages: dnf(8), rpm(8)